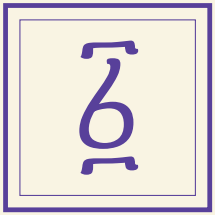
Business cases

Weekend 2: Representation learning

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026 𓂀𓋹𓂀𓆑



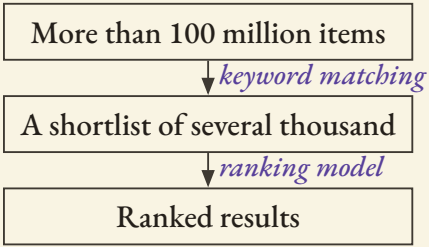
Section 1

Etsy

Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.

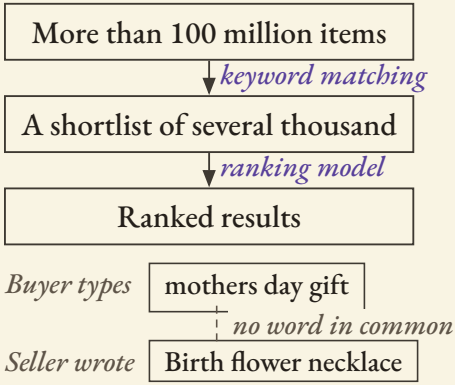


Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

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Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.



Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

The training data: a month of search logs

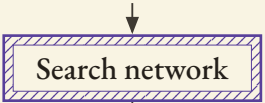
1 month: more than 150 million rows, from up to 30 million users.

Search	Who searched	Item	Action
mothers day gift	Zurich, searched necklaces	Flower necklace	purchase
bridge photos	New York, clicked print shops	Bridge print	cart add
wool socks	Oslo, searched knitting	Knitted socks	cart add

The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.

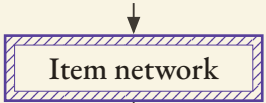
Every search and every item becomes a list of numbers

The search words and who searched



mothers day gift, in Zurich

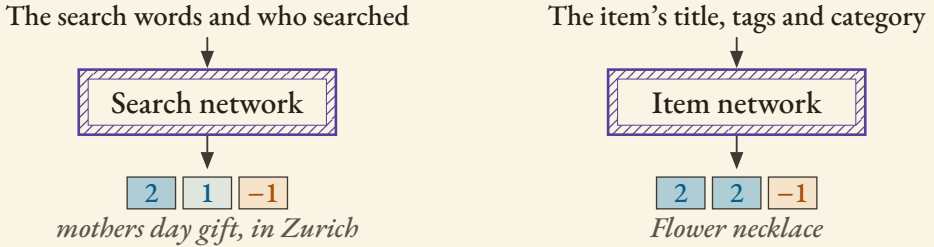
The item's title, tags and category



Flower necklace

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

Every search and every item becomes a list of numbers



Training makes the inner product high for items bought or added to the cart after the search: here 7.

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

A search arrives: sort by inner product

mothers day gift 2 1 −1

Invented numbers.

A search arrives: sort by inner product

mothers day gift	2	1	−1	*inner product*
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

Sorted: the top two join the shortlist

Flower necklace	7
Recipe board	4
Photo mug	3
Gift wrap	−4

Cost and value

Cost

72 hours *on 4 GPUs*

1 month *of search logs*

Value: A/B tests, relative change

+2.63% *site conversion rate*

+5.58% *search purchase rate*

Jha and others (Etsy), IEEE FMLDS 2024.

Cost and value

Cost

72 hours *on 4 GPUs*

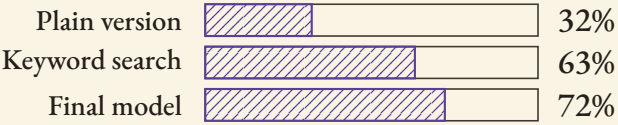
1 month *of search logs*

Value: A/B tests, relative change

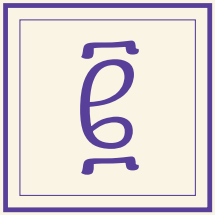
+2.63% *site conversion rate*

+5.58% *search purchase rate*

Popular searches: share of the bought products found in the top 100



Jha and others (Etsy), IEEE FMLDS 2024.



Section 2

Spotify

Spotify's audiobooks: little data to learn from

When Spotify launched audiobooks it had little data to learn from.

Each audiobook had to be bought before it could be streamed, and about a quarter of the audiobook listeners made three quarters of the listening hours.

De Nadai and others (Spotify), WWW 2024 Companion.

Spotify's audiobooks: little data to learn from

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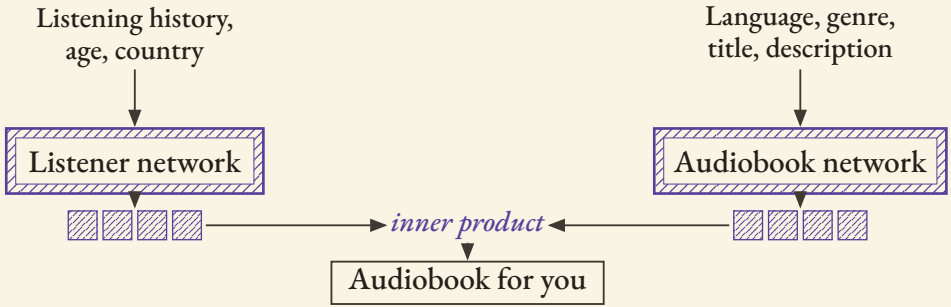
Over 70%

of the first audiobook listeners already listened to podcasts.

De Nadai and others (Spotify), WWW 2024 Companion.

128 numbers for every listener and every audiobook

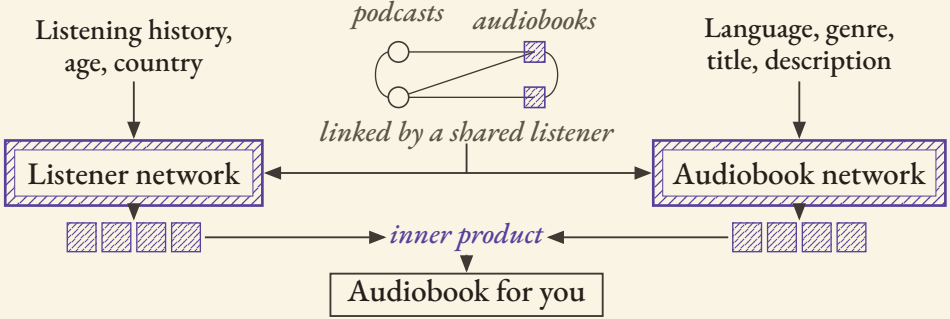
The matchmaker learned the numbers. Here two networks compute them.



De Nadai and others (Spotify), WWW 2024 Companion.

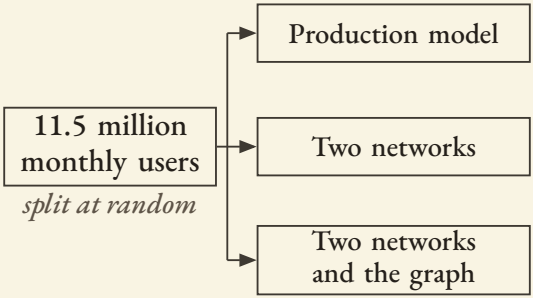
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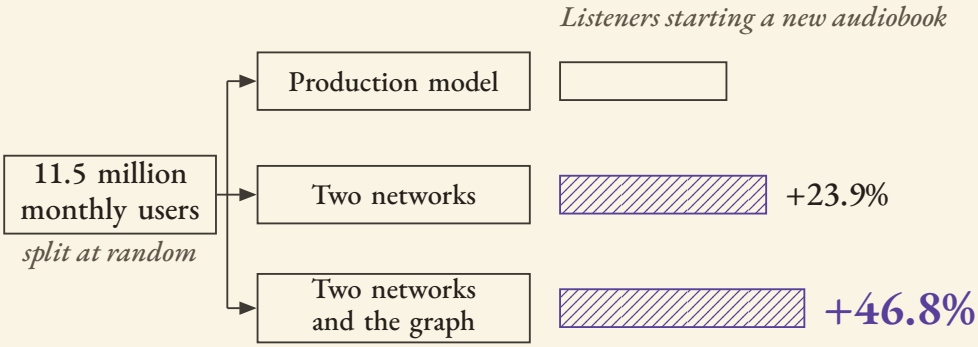
De Nadai and others (Spotify), WWW 2024 Companion.

The value: an A/B test

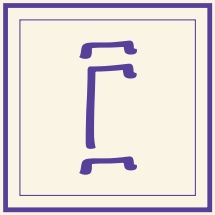


De Nadai and others (Spotify), WWW 2024 Companion.

The value: an A/B test



De Nadai and others (Spotify), WWW 2024 Companion.

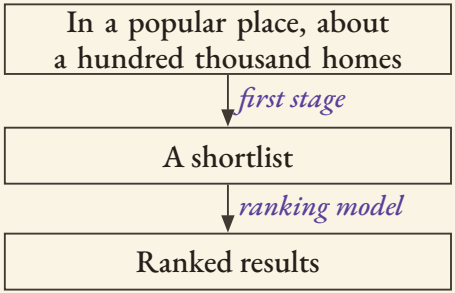


Section 3

Airbnb

Airbnb: the first stage of search

The first stage evaluates tens of thousands of homes.

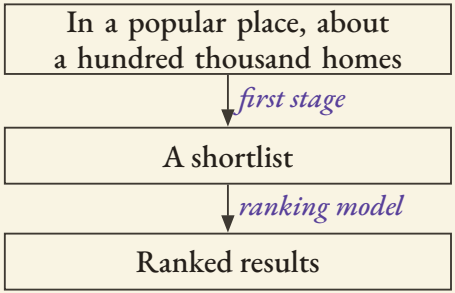


Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Airbnb: the first stage of search

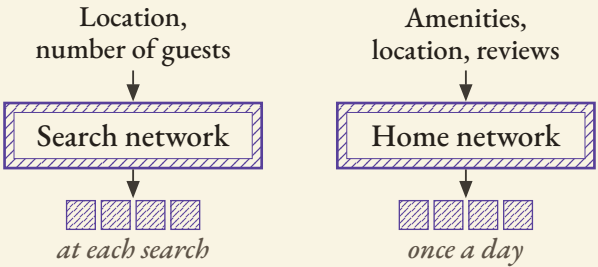
The first stage evaluates tens of thousands of homes.

It used to be only a simple linear model that scored every home once a day, whatever the search.



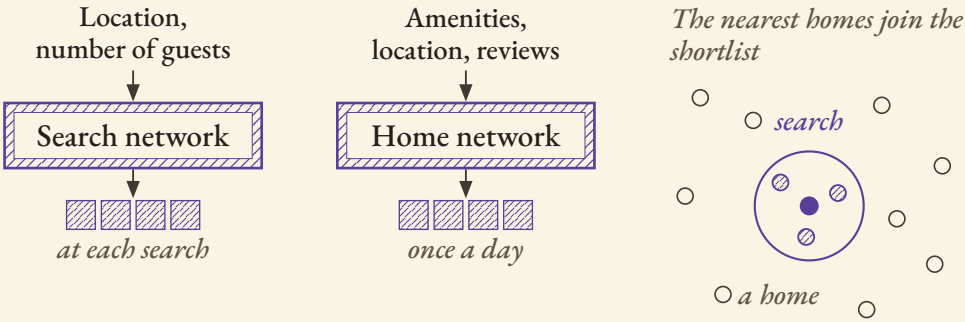
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes



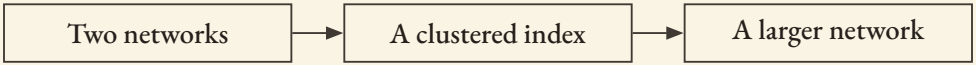
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes



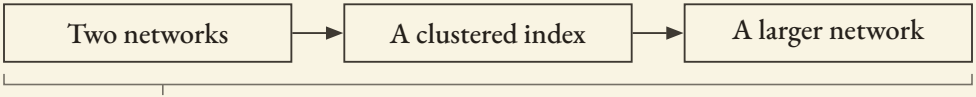
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



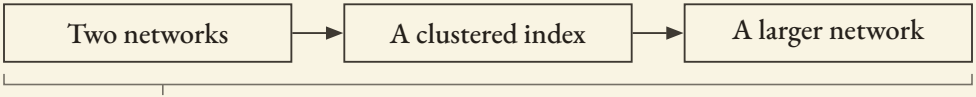
+0.31%

conversion, all three, relative change

Airbnb: “a large portion of the team’s annual goal”.

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.

The value: three A/B tests



+0.31%

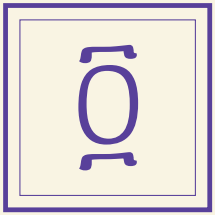
conversion, all three, relative change

Airbnb: “a large portion of the team’s annual goal”.

+2.3% *bookings from promotional emails*

−16% *compute with the clustered index*

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.



Section 4

Spotify’s home page

Spotify's home page: music or podcasts?

The home page shows shelves of music and podcasts. A common rule gives users the mix of their past listening.

But tastes can change with the hour, the day of the week, even the device.

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

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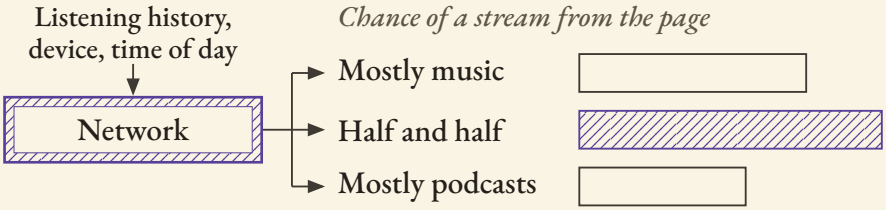
Over 600 million

users of the home page

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Epsilon-greedy on the home page

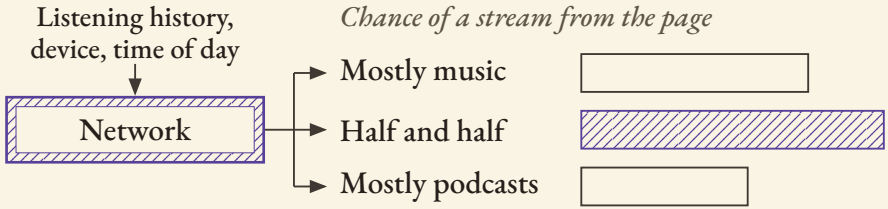
The mirror counted taps. Here a network estimates the chances.



Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Epsilon-greedy on the home page

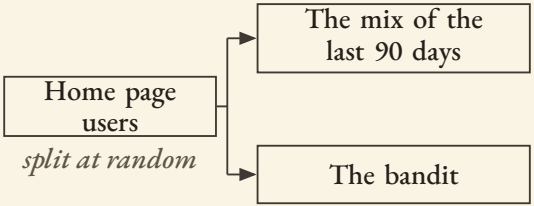
The mirror counted taps. Here a network estimates the chances.



98.5% of the pages: the highest estimate.
1.5%: a random mix near the usual one.

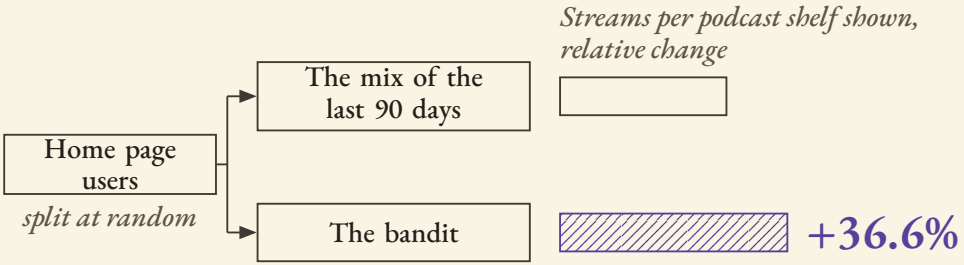
Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



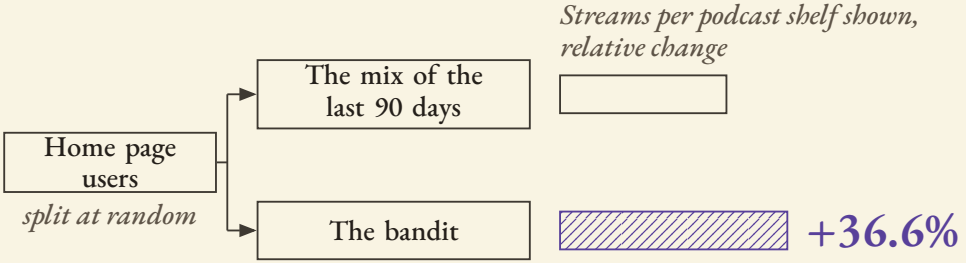
Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

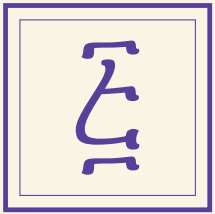
The value: an A/B test



In production as of March 2025.

+1.28% *minutes listened on Spotify, relative change*

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

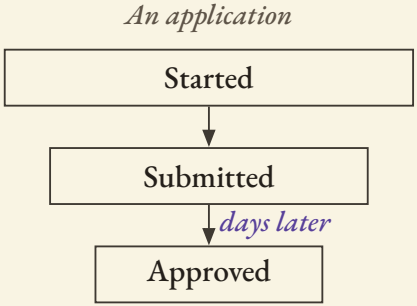


Section 5

Amazon Payments

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

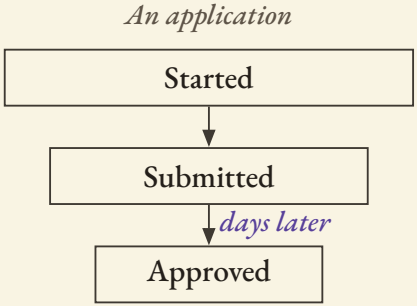


John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

A page that wins the first step can lose the last.



John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*
Pair A	6	3	1
Pair B	4	4	4
Pair C	3	2	3

The first step alone would choose pair A.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*	*Sum*
Pair A	6	3	1	10
Pair B	4	4	4	12
Pair C	3	2	3	8

The first step alone would choose pair A.
Amazon shows the highest sum: pair B.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers



Approvals rose by a high single-digit percentage.
relative change so far, not reported as significant

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

Amazon: “Thorough exploration can confirm that a content pool contains no winner.”

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

6

Section 6

LinkedIn's ads

LinkedIn's ads: the chance of a click

The chance that a member clicks an ad enters the ad auction, and the price depends on it.

Advertisers choose which clicks they pay for: some count likes or comments, others only visits to their website.

Borisyuk and others (LinkedIn), KDD 2024.

LinkedIn's ads: the chance of a click

The chance that a member clicks an ad enters the ad auction, and the price depends on it.

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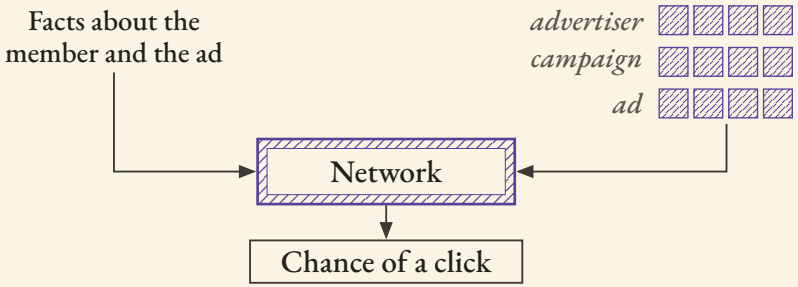
Over 1 billion

members

Borisyuk and others (LinkedIn), KDD 2024.

Numbers for every advertiser, campaign and ad

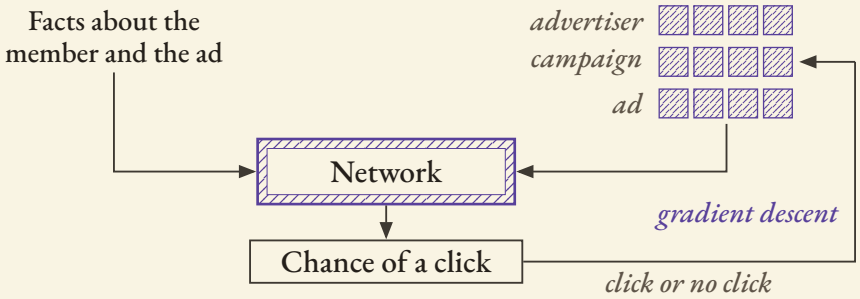
The matchmaker learned lists from ratings; here, from clicks.



Borisyuk and others (LinkedIn), KDD 2024.

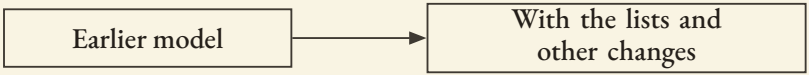
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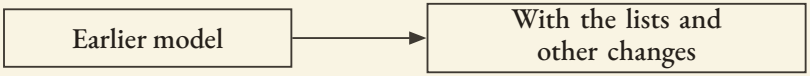
Borisyuk and others (LinkedIn), KDD 2024.

The value: A/B tests



Borisyuk and others (LinkedIn), KDD 2024.

The value: A/B tests



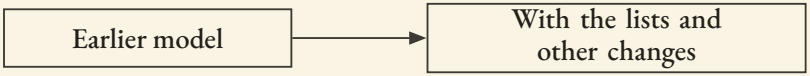
+4.3%

click rate on ads, relative change

LinkedIn put the new model into production.

Borisyuk and others (LinkedIn), KDD 2024.

The value: A/B tests



+4.3%

click rate on ads, relative change

LinkedIn put the new model into production.

−96%

training time in an offline test, when each new model builds on the earlier ones and learns from the newest half day of data

Borisyuk and others (LinkedIn), KDD 2024.